■ Complete Exam Study Guide

Object Oriented Programming Using Java - II

T.Y.B.Sc Computer Science | Internal Test (15 Marks)

Q.1 — One Mark Questions (Solve Any 5)

a) Interfaces Implemented by TreeSet

TreeSet implements: **Set, SortedSet, and NavigableSet** interfaces (also indirectly implements Iterable and Collection).

Definition: TreeSet stores elements in **sorted ascending order** using a balanced binary tree. It does **not allow duplicate elements**.

One-line answer: TreeSet implements Set, SortedSet, and NavigableSet interfaces.

b) ArrayList is a List Implemented Using a Resizable Array — Comment

- ArrayList is a class in **java.util** package that implements the **List interface**.
- Internally, it uses a **dynamic/resizable array** to store elements.
- When the array becomes full, Java **automatically increases the size** by creating a larger array and copying elements.
- It allows **duplicate elements** and maintains **insertion order**.
- Accessing elements is fast (by index), but **adding/removing in the middle is slow** because elements need to shift.
- **Conclusion:** Yes, ArrayList IS a resizable array — you don't declare size in advance; it grows and shrinks automatically.

c) Use of Collection Interface

Definition: The Collection interface is the **root interface** of the Java Collections Framework, in the **java.util** package.

Uses / Purpose:

- Provides basic operations: add, remove, search, and iterate on a group of objects.
- Defines common methods used by all collection classes (ArrayList, HashSet, TreeSet, etc.).
- Allows manipulation of a group of objects as a single unit.
- Helps achieve code reusability.

Key Methods: add(), remove(), contains(), size(), iterator(), clear()

d) Two Ways of Creating a Thread

Way 1: Extending the Thread class

```
class MyThread extends Thread {
   public void run() { System.out.println("Thread running"); }
}
// In main:
```

```
MyThread t = new MyThread();  t.start();
```

Way 2: Implementing the Runnable interface

```
class MyRunnable implements Runnable {
   public void run() { System.out.println("Thread running"); }
}
// In main:
Thread t = new Thread(new MyRunnable());  t.start();
```

Key Difference: Way 1 cannot extend another class. Way 2 (Runnable) is more flexible as the class can still extend another class.

e) Two Methods of Inter-Thread Communication

Definition: Inter-thread communication allows threads to cooperate/communicate while sharing resources.

Method	Description
wait()	Makes current thread pause and release the lock until notify() is called
notify()	Wakes up ONE waiting thread that called wait() on the same object
notifyAll()	Wakes up ALL waiting threads that called wait() on the same object

Exam answer (two methods): wait() and notify()

f) Difference Between Iterator and ListIterator

Feature	Iterator	ListIterator
Direction	Forward only	Forward AND Backward
Applicable to	Any Collection	Only List (ArrayList, LinkedList)
Can add elements?	No	Yes — add() method
Can replace elements?	No	Yes — set() method
Methods	hasNext(), next(), remove()	hasNext(), next(), hasPrevious(), previous(), add(), set()

Q.2 — 5 Mark Programs (Solve Any 2)

a) Store 'n' Employee Info in Hashtable and Display

```
import java.util.*;

class EmployeeHashtable {
  public static void main(String args[]) {
    Scanner sc = new Scanner(System.in);
    Hashtable<Integer, String> ht = new Hashtable<Integer, String>();

    System.out.print("Enter number of employees: ");
    int n = sc.nextInt();

    for (int i = 1; i <= n; i++) {
      System.out.print("Enter Employee ID: ");
      int id = sc.nextInt();
      System.out.print("Enter Employee Name: ");
```

```
      String name = sc.next();
      ht.put(id, name);
    }

    System.out.println("--- Employee Details ---");
    System.out.println("ID\tName");
    Set<Integer> keys = ht.keySet();
    for (Integer key : keys) {
      System.out.println(key + "\t" + ht.get(key));
    }
    sc.close();
  }
}
```

Output: Enter number of employees: 3 | Enter Employee ID: 101 | Enter Employee Name: Rahul ...
--- Employee Details ---
ID Name
103 Amit 102 Priya 101 Rahul

b) Display 'examination' 50 Times Using Multithreading

```
class PrintThread extends Thread {
  public void run() {
    for (int i = 1; i <= 50; i++) {
      System.out.println(i + ": examination");
    }
  }
}

class MultiThreadDemo {
  public static void main(String args[]) {
    PrintThread t = new PrintThread();
    t.start();
  }
}
```

Output: 1: examination 2: examination ... 50: examination

c) LinkedList of 4 Integers — Add at First + Delete Last

```
import java.util.*;

class LinkedListDemo {
  public static void main(String args[]) {
    LinkedList<Integer> ll = new LinkedList<Integer>();
    ll.add(10);
    ll.add(20);
    ll.add(30);
    ll.add(40);

    System.out.println("Original List: " + ll);

    // I] Add element at first position
    ll.addFirst(5);
    System.out.println("After adding 5 at first: " + ll);

    // II] Delete last element
```

```
    ll.removeLast();
    System.out.println("After deleting last element: " + ll);
  }
}
```

Output:
Original List: [10, 20, 30, 40]
After adding 5 at first: [5, 10, 20, 30, 40]
After deleting last element: [5, 10, 20, 30]

Quick Revision Table

Topic	Key Point
TreeSet interfaces	Set, SortedSet, NavigableSet
ArrayList	Resizable array, allows duplicates, maintains insertion order
Collection Interface	Root interface; provides add, remove, iterate, size methods
Thread creation	1. Extend Thread class 2. Implement Runnable interface
Inter-thread methods	wait(), notify(), notifyAll() — defined in Object class
Iterator vs ListIterator	Iterator = forward only; ListIterator = both directions
Hashtable	Stores key-value pairs; use put() to insert, get() to retrieve
LinkedList methods	addFirst(), removeLast(), addLast(), removeFirst()

Key Definitions (One-Liners)

Thread: A lightweight sub-process; the smallest unit of processing.

Multithreading: Running two or more threads simultaneously in a single program.

Collection: A group of objects stored and manipulated as a single unit.

TreeSet: A Set that stores elements in sorted ascending order using a tree.

ArrayList: A List backed by a dynamic resizable array.

Hashtable: Stores data as key-value pairs; thread-safe; no null keys.

LinkedList: A doubly-linked list allowing add/remove at any position.

Iterator: Used to loop through a collection in the forward direction only.

ListIterator: Like Iterator, but works in both directions on a List.

All the Best for Your Exam! ■